

Tasbir Kabir

Digital HQ Platform

Complete Platform Documentation

Metric	Value
Framework	Next.js 16 (App Router) + TypeScript
Database	Prisma ORM + SQLite
Styling	Tailwind CSS 4 + shadcn/ui
Ebooks	8 published with premium reader
Blog Posts	3 published with markdown
Free Resources	8 downloadable
JSON-LD Schemas	12 types across all pages
Admin Panel	11 sections (CMS + security)
Payment Gateways	SSL Commerz, Stripe, UddoktaPay
AI Discovery	llms.txt, ai.txt, robots.txt

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1. Platform Features

1.1 Ebook Store & Premium Reader

The platform includes 8 published ebooks with full chapter content, CSS-designed covers, and a premium web-based reader. The reader features a Dynamic Island table of contents, reading progress tracking, bookmarks, highlights, sepia/dark/light themes, and a preview system that allows free reading of the first chapters before a paywall. Each book has a dedicated sales page with pricing, table of contents, reader reviews, FAQ, and a glassmorphism checkout modal supporting bKash, Nagad, Google Pay, and card payments.

1.2 Full Content Management System (CMS)

The admin panel includes a 6-tab Site Settings editor that allows the site owner to manage nearly every piece of content without touching code:

- **Homepage tab** — Hero section (eyebrow, title, subtitle, CTAs, image), section labels, editable Stats list, and What I Build cards list.
- **About tab** — Hero, mission, services, process, timeline, achievements, and CTA — all editable with multi-paragraph support.
- **Footer tab** — Bio, copyright, tagline, and fully editable footer columns with nested link management.
- **Navigation tab** — Add, edit, remove, and reorder nav links.
- **Branding tab** — Brand name, tagline, logo, favicon, OG image, accent color, and all social links.
- **Contact & SEO tab** — Contact email, phone, default SEO title and description.

1.3 Admin Account & Security

The admin has a dedicated Account panel and Security panel for full account management:

- **Profile management** — Edit name, email, and profile image.
- **Secure password change** — Verifies current password, enforces strong-password policy (8+ chars, uppercase, lowercase, number), hashes with bcrypt (12 rounds), logs the change, and invalidates all other sessions.
- **Active sessions** — View every device with IP, last-active time, and one-click device logout (DB-backed session revocation).
- **Login history** — Last 30 login attempts with success/fail status and reason.
- **Two-Factor Authentication (2FA)** — TOTP-based 2FA with QR code setup, verify-to-enable, and verify-to-disable. Enforced at login with a 6-digit code step in the auth modal.

1.4 Blog & Resources

The blog includes a markdown-based editor in the admin panel with category, tags, cover image, and SEO fields. Blog posts feature a Dynamic Island TOC, drop caps, reading time, and related content. The resources section offers 8 free downloads (prompt packs, checklists, templates) with category filtering and download tracking. Both blog and resources have full CRUD in the admin.

2. SEO, GEO & AEO Optimization

2.1 Structured Data (JSON-LD Schema)

The platform implements 12 JSON-LD schema types across all pages, helping search engines and AI answer engines understand and cite the site content:

Schema Type	Pages	Purpose
Person	Home, About	Core identity markup for the personal brand
Organization	Home	Business entity with contact point
WebSite	Home	Site info with SearchAction for sitelinks
ProfilePage	Home	Helps AI engines identify the person
AboutPage	About	About page entity with mainEntity
ContactPage	Contact	Contact page with email/phone
Book	Book sales pages	Ebook with offers and ratings
Product	Book sales pages	Digital product with pricing
Article	Blog posts	Article with author, date, keywords
BreadcrumbList	Book, Blog post	Navigation context
FAQPage	About, Resources, Book, Knowledge Hub	FAQ results + AI citation
CollectionPage	Books, Knowledge Hub	Collection of content items

2.2 FAQ Sections (GEO)

Four FAQ datasets with 24 total Q&A; pairs are deployed across the About, Resources, Book, and Knowledge Hub pages. Each FAQ section renders with an accordion UI and injects FAQPage JSON-LD schema, earning rich results in search engines and helping AI answer engines cite the site content directly.

2.3 Knowledge Hub (GEO Content Clusters)

The Knowledge Hub is a GEO-optimized content aggregation page that groups all books, blog posts, and resources into 5 topic clusters: AI Agents, Automation Systems, Web Development, Marketing & Media Buying, and Business Systems. Each cluster links to relevant content, helping AI engines discover topical relationships. The page includes CollectionPage and FAQPage JSON-LD.

2.4 AEO (Answer Engine Optimization)

The platform implements full AEO infrastructure to help AI answer engines (ChatGPT, Claude, Perplexity, Gemini) discover, understand, and cite the site content:

- **llms.txt** — A markdown-formatted AI discovery file (<https://llmstxt.org>) that summarizes the site purpose, services, key content, and contact info in a format optimized for LLM consumption.
- **ai.txt** — A JSON-formatted AI crawler discovery file declaring which crawlers are welcome, what content they may use, and providing machine-readable site entities.
- **robots.txt** — Explicitly allows all major AI crawlers (GPTBot, Claude-Web, PerplexityBot, Google-Extended, CCBot, Applebot-Extended, cohere-ai, and more) with dedicated Allow rules.
- **Enriched Person schema** — Added nationality, knowsAbout (expanded with OpenAI, Claude, Next.js, WordPress, ChatGPT), knowsLanguage, and award fields for richer AI entity recognition.
- **Canonical URLs** — Every page has a canonical link tag to prevent duplicate content issues.
- **Rich metadata** — OpenGraph with locale and siteName, Twitter cards, robots directives with max-image-preview:large and max-snippet:-1.

3. Performance Optimizations

3.1 Image Optimization

All images have been converted to WebP format, reducing file sizes by up to 93%. The hero image (previously 533KB as PNG) is now 38KB as WebP. All image references auto-normalize from .png to .webp. The hero image uses `fetchPriority="high"` and `loading="eager"` as the LCP element, while non-critical images use `loading="lazy"` with explicit width/height attributes to prevent layout shift.

3.2 JavaScript Bundle Reduction

Heavy components are lazy-loaded via `React.lazy()` and `Suspense`, cutting approximately 257 KiB of unused JavaScript from the initial bundle. The following components load on-demand only when needed:

- **AdminView** — Entire admin panel (only loads when admin navigates to /admin)
- **EbookReader** — Premium reader (only loads when user opens a book)
- **CheckoutModal** — Payment flow (only loads when user clicks Buy)
- **LibraryView, AccountView, SearchView, KnowledgeHubView** — Secondary views loaded on navigation

3.3 Font & Render-Blocking Optimization

All Google Fonts use `display: "swap"` so text renders immediately with fallback fonts. The Geist Mono font (used only in admin) and Newsreader font (used only in the reader) have `preload: false` to prevent them from blocking initial render. This eliminates approximately 450ms of render-blocking time.

3.4 Metadata & SEO Tags

The layout includes `metadataBase` for resolving relative OG image URLs, canonical link tags on every page, robots directives with `max-image-preview:large` and `max-snippet:-1` for rich search results, and expanded keywords including OpenAI, Claude, Make.com, lead generation, and business automation.

4. Security Implementation

4.1 Authentication & Session Management

The platform uses HMAC-SHA256 signed session tokens backed by a database. Each session token carries a unique session ID (sid) that is looked up in the database on every request, enabling real-time session revocation. Sessions expire after 7 days and are automatically cleaned up.

4.2 Password Security

Passwords are hashed with bcrypt using 12 salt rounds. The password change flow verifies the current password first, enforces a strong-password policy (8+ characters, uppercase, lowercase, number), hashes the new password, logs the change in the audit log, and invalidates all other sessions except the current one. Password reset tokens are time-limited and single-use.

4.3 Rate Limiting & Input Validation

All API endpoints use per-IP rate limiting (10 login attempts per 15 minutes, 5 registrations per hour, 12 password changes per hour). All inputs are validated with Zod schemas server-side. Error responses are sanitized to never expose internal details.

4.4 Audit Logging

Every admin action is logged with the user, action, target, and timestamp. Security events (failed logins, password changes, 2FA events) are logged with IP and email. Login history records every authentication attempt with success/fail status and reason.

4.5 Two-Factor Authentication

TOTP-based 2FA is implemented using otplib. The admin can enable 2FA from the Security panel by scanning a QR code with Google Authenticator, Authy, or 1Password. When 2FA is enabled, login requires a 6-digit code after password verification. The 2FA flow uses a 5-minute signed pending token to carry the user ID through the challenge.

5. Deployment Guide

5.1 Environment Variables

The following environment variables must be set in the Vercel project settings:

Variable	Description
DATABASE_URL	SQLite database path (file:./db/custom.db)
SESSION_SECRET	HMAC secret for signing session tokens (32+ random chars)
NEXTAUTH_SECRET	Secret for NextAuth (if used)
SSL_COMMERZ_STORE_ID	SSL Commerz store ID (optional)
SSL_COMMERZ_STORE_PASSWD	SSL Commerz store password (optional)
STRIPE_SECRET_KEY	Stripe API secret key (optional)
RESEND_API_KEY	Resend email API key (optional)

5.2 Vercel Deployment Steps

1. Push the code to a GitHub repository.
2. Go to vercel.com and import the repository.
3. Set all environment variables in the Vercel project settings.
4. Deploy. Vercel will automatically run next build and deploy.
5. Set up a SQLite database (Vercel Postgres or external) and update DATABASE_URL.
6. Run `bun run db:push` to create the database schema.
7. Run the seed script to populate initial content.
8. Set up a custom domain (tasbirkabir.site) in Vercel settings.
9. Update the DNS records to point to Vercel.

5.3 Admin Access

The default admin account is **admin@tasbirkabir.site** with password **ChangeMe2026!**. Change this password immediately after deployment from the Admin > Account panel. Enable 2FA from the Admin > Security panel for maximum security.

5.4 Post-Deployment Checklist

- Change the admin password from the default.

- Enable 2FA on the admin account.
- Update all social links and contact info in Site Settings > Branding.
- Verify the sitemap.xml and rss.xml are accessible.
- Submit the sitemap to Google Search Console.
- Test the llms.txt and ai.txt endpoints.
- Run a PageSpeed Insights test to verify performance.
- Run an AEO audit to verify AI readiness.
- Set up email forwarding for the contact form.
- Configure payment gateway credentials if selling ebooks.

6. Technical Architecture

6.1 Technology Stack

Layer	Technology
Framework	Next.js 16 with App Router
Language	TypeScript 5 (strict mode)
Styling	Tailwind CSS 4 + shadcn/ui (New York)
Database	Prisma ORM + SQLite
Auth	HMAC-signed sessions + bcrypt + TOTP 2FA
State	Zustand (client) + useData hook (server)
Animations	Framer Motion (motion/react)
Icons	Lucide React
Fonts	Geist, Instrument Serif, Newsreader
Payments	SSL Commerz, Stripe, UddoktaPay
Email	Resend (password reset, broadcasts)
SEO	12 JSON-LD schemas, sitemap, RSS, robots, llms.txt

6.2 Database Models

The Prisma schema includes 14 models: Book, Asset, Resource, BlogPost, Testimonial, User, Session, LoginHistory, LibraryAccess, ReadingProgress, Order, NewsletterSub, Broadcast, ContactMessage, SiteSettings, and AnalyticsEvent. The SiteSettings model is a singleton with 40+ fields covering all editable content including homepage sections, about page, footer, navigation, branding, social links, contact info, and SEO defaults.

6.3 Routing Architecture

The platform uses a single / route with client-side view routing via a Zustand navigation store. All views (home, about, books, book, resources, blog, post, contact, library, account, search, knowledge, admin) are rendered client-side. API routes under /api/* handle all server logic including auth, content delivery, admin operations, and payments.

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